

Surface Areas and Volumes

Surface areas and volumes of combinations of two basic solids, with formulae, visualisation tips and worked examples.

Introduction

In Class 9 you found the surface area and volume of single solids such as a cuboid, cylinder, cone and sphere. In real life, many objects are not a single solid, a capsule is a cylinder with a hemisphere at each end, a toy may be a cone on a hemisphere, and a hut may be a cylinder topped by a cone. This chapter is about **combination solids**: solids made by joining two or more basic solids. The two ideas to keep in mind are very simple, when solids are joined, some faces get hidden, so the total surface area is not just the sum of the individual surface areas, but the total volume is just the sum of the individual volumes.

The Basic Solids, A Quick Recall

All combination problems use the formulas you already know. Keep these handy before you start combining solids. Here, l = length, b = breadth, h = height, r = radius, a = side of cube, and s or l_c = slant height of cone.

Formulas of basic solids

Cuboid: CSA (lateral) = $2h(l + b)$, TSA = $2(lb + bh + hl)$, Volume = $l \times b \times h$

Cube (side a): CSA = $4a^2$, TSA = $6a^2$, Volume = a^3

Cylinder: CSA = $2\pi rh$, TSA = $2\pi r(r + h)$, Volume = $\pi r^2 h$

Cone: slant height $s = \sqrt{r^2 + h^2}$, CSA = πrs , TSA = $\pi r(s + r)$, Volume = $\frac{1}{3} \pi r^2 h$

Sphere: Surface Area = $4\pi r^2$, Volume = $\frac{4}{3} \pi r^3$

Hemisphere: CSA = $2\pi r^2$, TSA = $3\pi r^2$, Volume = $\frac{2}{3} \pi r^3$

Use $\pi = 22/7$ if the radius is a multiple of 7, otherwise use 3.14 or leave the answer in terms of π , as the question says. Most CBSE questions deliberately give radii like 7 cm, 14 cm or 3.5 cm so that 22/7 cancels neatly.

Surface Area of a Combination of Solids

When two solids are joined, the part where they touch is no longer on the outside. So you must **not** add the total surface areas of the two solids. Instead, you add only the **curved (or lateral) surface areas that remain exposed**.

Key idea, Surface area of a combination

Total Surface Area of the combination = sum of the curved / exposed surface areas of each part.

The area of the common circular face (or joining face) is counted in neither solid, because it becomes internal.

For example, if a cone of radius r and slant height s is placed on a hemisphere of the same radius r , the flat circular base of each solid is hidden. The total surface area you can see is:

$$\text{TSA} = \text{CSA of cone} + \text{CSA of hemisphere} = \pi rs + 2\pi r^2$$

Similarly, a capsule (cylinder with a hemisphere at each end) of radius r and cylinder height h has:

$$\text{TSA} = \text{CSA of cylinder} + 2 \times \text{CSA of hemisphere} = 2\pi rh + 4\pi r^2$$

Worked Example 1, Surface area

A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find its total surface area. (Take $\pi = 22/7$)

Radius $r = 3.5$ cm. Height of cone = total height – radius of hemisphere = $15.5 - 3.5 = 12$ cm.

Slant height of cone, $s = \sqrt{(r^2 + h^2)} = \sqrt{(3.5^2 + 12^2)} = \sqrt{(12.25 + 144)} = \sqrt{156.25} = 12.5$ cm.

CSA of cone = $\pi rs = (22/7) \times 3.5 \times 12.5 = 22 \times 0.5 \times 12.5 = 137.5 \text{ cm}^2$.

CSA of hemisphere = $2\pi r^2 = 2 \times (22/7) \times 3.5 \times 3.5 = 2 \times 22 \times 0.5 \times 3.5 = 77 \text{ cm}^2$.

Total surface area of toy = $137.5 + 77 = \mathbf{214.5 \text{ cm}^2}$.

Note: we did not add the base area πr^2 of either solid, because that circle is hidden where the two parts are joined.

Volume of a Combination of Solids

Volume is simpler than surface area. Matter fills the whole object, so even the internal joining face counts as occupied space. The volume of the combination is just the **sum** of the volumes of the individual solids.

Key idea, Volume of a combination

Volume of the combination = sum of the volumes of the parts that make it up.

For the same toy above (cone + hemisphere of radius r , cone height h):

$$\text{Volume} = \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3 = \frac{1}{3} \pi r^2 (h + 2r)$$

Worked Example 2, Volume

A solid is composed of a cylinder with hemispherical ends. The radius of the hemispherical ends is 14 cm and the total length of the solid is 50 cm. Find its volume. (Take $\pi = 22/7$)

Radius $r = 14$ cm. The two hemispherical ends together make one full sphere.
Height of the cylindrical part = total length $- 2r = 50 - 28 = 22$ cm.

Volume of cylinder = $\pi r^2 h = (22/7) \times 14 \times 14 \times 22 = 22 \times 28 \times 22 = 13552$ cm^3 .

Volume of two hemispheres (= one sphere) = $\frac{4}{3} \pi r^3 = (\frac{4}{3}) \times (22/7) \times 14^3 = (4 \times 22 \times 2744)/21 = 11498.67 \text{ cm}^3 \approx 34496/3 \text{ cm}^3$.

More neatly: Volume = $13552 + 11498.67 = \mathbf{25050.67 \text{ cm}^3}$ (or exactly $751520/30 \text{ cm}^3$). Using fraction: cylinder = $13552 = 40656/3$, sphere = $34496/3$, sum = $\mathbf{75152/3 \text{ cm}^3 \approx 25050.67 \text{ cm}^3}$.

Tip: when you see "cylinder with hemispherical ends", always split it as cylinder + sphere for volume, but as cylinder CSA + sphere surface area for surface area.

A very common exam question is a "capsule", "test tube" (cylinder + hemisphere), "toy" (cone + hemisphere), or a "hut" (cylinder or cube + cone/hemisphere). The method is always the same: identify the parts, use the same radius for all parts that are joined, and then add the right areas or volumes.

Summary

- **Combination solids** are made by joining two or more basic solids along a common face.
- **Surface area of a combination** = sum of the exposed curved surface areas. Do not add the area of the internal joining face.
- **Volume of a combination** = sum of the volumes of all the parts.
- Basic formulas: Cylinder CSA $2\pi rh$, Cone CSA πrs , Sphere $4\pi r^2$, Hemisphere CSA $2\pi r^2$; volumes $\pi r^2 h$, $\frac{1}{3}\pi r^2 h$, $\frac{4}{3}\pi r^3$, $\frac{2}{3}\pi r^3$.
- Always check that all parts share the same radius where they are joined, and use $\pi = 22/7$ when radii are multiples of 7.

Practice Questions

1. A solid is in the form of a cylinder with a hemisphere mounted on one end. The radius is 7 cm and the height of the cylinder is 20 cm. Find the total surface area of the solid. (Take $\pi = 22/7$)

2. A toy is made by mounting a cone of height 24 cm on a hemisphere of radius 7 cm. Find the volume of the toy.

3. A capsule is in the form of a cylinder with two hemispherical ends. The total length of the capsule is 14 cm and the diameter is 7 cm. Find its surface area and volume.

4. A solid consists of a circular cylinder with exact height 120 cm and radius 60 cm, topped by a cone of the same radius and height 36 cm. Find its total volume.